1. Introduction

The National Mutual Economy (NME) Report is one of the main ways policymakers, regulators and the public understand Australia's co-operative and mutual enterprises (CMEs). Its Top 100 ranks organisations by turnover, so it mainly shows size. Karl decides what the report measures. This dashboard gives him evidence for going further: which sectors the ranking leaves out, where its largest co-ops are based, and whether their financial profiles still fit the community purpose the sector was built on.

This matters now because the sector is consolidating. CEMI records 1,000 inactive CMEs, including 45 that have demutualised, and financial institutions are growing partly to meet Basel III requirements (Mazzarol, 2026). The co-ops that survive are getting bigger, and the open question is whether they are also moving further from the communities they were created to serve.

International cases show why this is worth measuring. Mondragon's flagship co-operative, Fagor Electrodomésticos, became a multinational competing in the global appliance market and closed in 2013, which raised questions about the viability of the co-operative model itself (Errasti et al., 2017). In the United States, REI reports 25 million members and customers and US$3.53 billion in 2024 revenue (REI Co-op, 2025), which raises the question of how a co-op stays distinct from its competitors at that scale. If Karl could also measure mission alignment, the NME Report could become a diagnostic tool for the health of the co-operative movement, as well as a leaderboard.

2. Design Template: Persona 2

Persona	'CME Data Analyst Karl'
Role	BCCM Data Analyst
Context of their work/role	Karl is in charge of the annual BCCM NME Report and works across its full life cycle, from data harvesting and cleansing to producing the report. The report is widely used to understand Australia's CME sector, so what Karl chooses to measure shapes how the sector is seen.
Key activities/decision-making needs and interests	Karl wants a visual data exploration environment for the annual NME reports, serving the different stakeholders interested in the sector. He needs evidence to decide which sectors the report should make visible, whether it should add a community lens, and whether a mission alignment measure is worth piloting.
DV Questions	Question 1: Which sectors show the largest gap between active CMEs and Top 100 representation? Question 2: Are Australia's largest co-ops drifting from the communities that need them most? Question 3: How far is success pulling Australia's biggest co-ops away from the people who need them most?

3. Data Set: Persona 2

3.1 Data sources

The Persona 2 data set covers 100 organisations and 28 sectors for FY2024-25, well above the minimum record requirement. It combines three public sources with fields derived by the team.

Source	Description
CEMI Discussion Paper DP2601 (Mazzarol, 2026)	1,810 active CMEs across 28 sectors (Table 1); Top 100 turnover, EBIT and assets (Appendix A)
BCCM NME Report 2026	Number of Top 100 co-ops in each industry; used to fact-check Top 100 names and values
ABS SEIFA 2021 (IRSAD) by Local Government Area	Socio-economic score and decile for each head office area
Derived by the team	Head office matched to its Local Government Area; Mission Drift Index calculated from profit margin, assets to turnover and SEIFA decile

3.2 Data preparation

Collection and checking. Top 100 names, turnover, EBIT and assets were extracted from Appendix A of the CEMI paper, and the 28-sector counts from its Table 1. Every record was then fact-checked against the source documents, including the NME Report 2026.

Linking location. Each head office was matched to its Local Government Area, which was joined to the ABS SEIFA score and decile for that area. Each co-op was also labelled Metro or Regional.

Calculated fields. EBIT margin (EBIT / turnover) and assets to turnover were calculated for each co-op. Each of the three MDI components was converted to a z-score, rescaled to a 0 to 1 range and averaged to give the MDI score.

Missing values. Two records have values missing from the source. These cells were left blank instead of being estimated, and the MDI for those two records averages only the components available.

Final file. The cleaned data was saved as Persona2_Karl_Dataset_FINAL.xlsx with three sheets: Metadata, which defines every field, Top100 and Sectors.

3.3 Sample data

The screenshots below show the Metadata sheet, the first 11 of the 100 records in the Top100 sheet (main columns shown), and the first 10 of the 28 rows in the Sectors sheet.

Persona 2 'CME Data Analyst Karl' dataset: metadata

Unit: BUSA3021, Session 2 2026. Client: BCCM. Reference year: FY2024-25 (single year). Records: 100 organisations (Top100) and 28 sectors (Sectors). Excludes member-owned superannuation funds.

Sources: NME Report 2026 (BCCM, CC BY 4.0); CEMI DP2601 (Mazzarol, 2026), Table 1 and Appendix A; ABS SEIFA 2021 IRSAD by LGA. HQ-to-LGA matching and MDI fields are derived by the project team.

Sheet	Field	Type / unit	Definition and source
Top100	Rank	Integer	Rank by gross turnover, FY2024-25 (NME 2026; CEMI Appendix A)
Top100	Co-op Name	Text	Organisation name, shortened from source where long
Top100	State	Text	State or territory of the organisation in source
Top100	Sector	Text	One of 8 project groupings of the source industry labels. NOT the same as the 28 CEMI sectors
Top100	Turnover ($M)	AUD millions	Gross annual turnover (source)
Top100	Sector Active CMEs	Integer	Active CMEs for the matching CEMI sector. Civic Risk Mutual has no match (recorded as 0)
Top100	Sector in Top 100	Integer	Top 100 count for that project grouping
Top100	Sector Gap	Integer	Sector Active CMEs minus Sector in Top 100. Not used in the dashboard; use the Sectors sheet
Top100	LGA (HQ)	Text	Local Government Area of the head office, matched by the project team
Top100	SEIFA Score	Index	ABS IRSAD 2021 score for the HQ LGA. ABS standardises scores to mean 1,000 and SD 100 at SA1 level
Top100	SEIFA Decile	1-10	ABS IRSAD decile for the HQ LGA (1 = most disadvantaged, 10 = most advantaged)
Top100	Location	Metro / Regional	Project classification of the HQ LGA
Top100	EBIT Margin	Ratio	EBIT / gross turnover. Blank for 2 records with no EBIT in source
Top100	Assets to Turnover	Ratio	Gross assets / gross turnover. Blank for 1 record with no assets in source
Top100	MDI Profit	0-1	(z + 3) / 6 of EBIT Margin, capped to 0-1, population SD
Top100	MDI Capital	0-1	(z + 3) / 6 of Assets to Turnover, capped to 0-1. The Capital Hoarding Ratio component
Top100	MDI Community	0-1	(z + 3) / 6 of SEIFA Decile, capped to 0-1
Top100	MDI Score	0-1	Unweighted mean of available components. A screening signal, not a verdict. Two records are partial: Tyrepower Group (1 component), NSW Sugar Milling (2)
Sectors	Sector	Text	One of 28 CEMI sectors (CEMI DP2601, Table 1)
Sectors	Active CMEs	Integer	Active CMEs in the sector (CEMI DP2601, Table 1)
Sectors	In Top 100	Integer	Organisations in the Top 100 by sector (NME Report 2026)
Sectors	Gap	Integer	Active CMEs minus In Top 100
Sectors	Pct Represented	Ratio	In Top 100 / Active CMEs
Sectors	Sector Category	Text	Project grouping of the 28 sectors into 6 categories

Rank	Co-op Name	State	Sector	Turnover ($M)	LGA (HQ)	SEIFA Score	SEIFA Decile	Location	EBIT Margin	Assets to Turnover	MDI Score
1	Co-operative Bulk Handling Ltd	WA	Agribusiness	5,829.6	Perth	1081.0	10	Metro	0.049	0.740	0.483
2	Hospital Contribution Fund (HCF)	NSW	Health Insurance	4,433.6	Sydney	1125.8	10	Metro	0.045	0.763	0.481
3	Capricorn Society Ltd	WA	Wholesale & Purchasing	3,974.1	Perth	1081.0	10	Metro	0.016	0.235	0.461
4	Australian Unity	VIC	Health Insurance	2,602.9	Melbourne	1085.5	10	Metro	0.072	4.035	0.518
5	HBF Health	WA	Health Insurance	2,424.1	Perth	1081.0	10	Metro	0.056	0.839	0.487
6	RACQ	QLD	Motoring	2,396.8	Brisbane	1060.7	9	Metro	0.069	2.482	0.484
7	RAC WA	WA	Motoring	1,906.3	Perth	1081.0	10	Metro	0.199	1.568	0.569
8	Heritage and Peoples Choice	QLD	Financial Services	1,387.1	Toowoomba	958.7	6	Regional	0.043	18.261	0.514
9	NGM Group	NSW	Financial Services	1,178.9	Newcastle	1012.1	9	Regional	0.141	19.611	0.643
10	Credit Union Australia (CUA)	QLD	Financial Services	1,095.4	Brisbane	1060.7	9	Metro	0.055	19.287	0.594
11	Teachers Health Fund	NSW	Health Insurance	1,080.8	Sydney	1125.8	10	Metro	0.018	0.797	0.466

Sector	Active CMEs	In Top 100	Gap	Pct Represented	Sector Category
Housing	232	0	232	0.0%	Social/Community
Sport & Recreation	220	1	219	0.5%	Social/Community
Community Services	168	1	167	0.6%	Social/Community
Medical Services	159	5	154	3.1%	Professional
Retailing	129	3	126	2.3%	Commercial
Agribusiness	126	11	115	8.7%	Primary Industry
Banking & Financial Services	118	39	79	33.1%	Financial
Education, Training, Childcare	112	2	110	1.8%	Social/Community
Arts & Culture	85	0	85	0.0%	Social/Community
Utilities	62	2	60	3.2%	Infrastructure

Figure 1: Sample of the Excel data set: Metadata sheet, Top100 records and Sectors records.

4.2 Visual Answers: Persona 2 'CME Data Analyst Karl'

The three questions below are designed to escalate. Q1 identifies what is missing from the NME Report. Q2 shows where the problem is concentrated. Q3 quantifies how bad it is. Together, they tell a single story: the co-operative sector's success metrics are hiding its biggest vulnerability.

All three answers sit on one Tableau dashboard, so Karl can move from the sector gap to location to the index without changing tools. He can hover over any mark to see the underlying values and use the colour legends to highlight groups such as Metro and Regional co-ops.

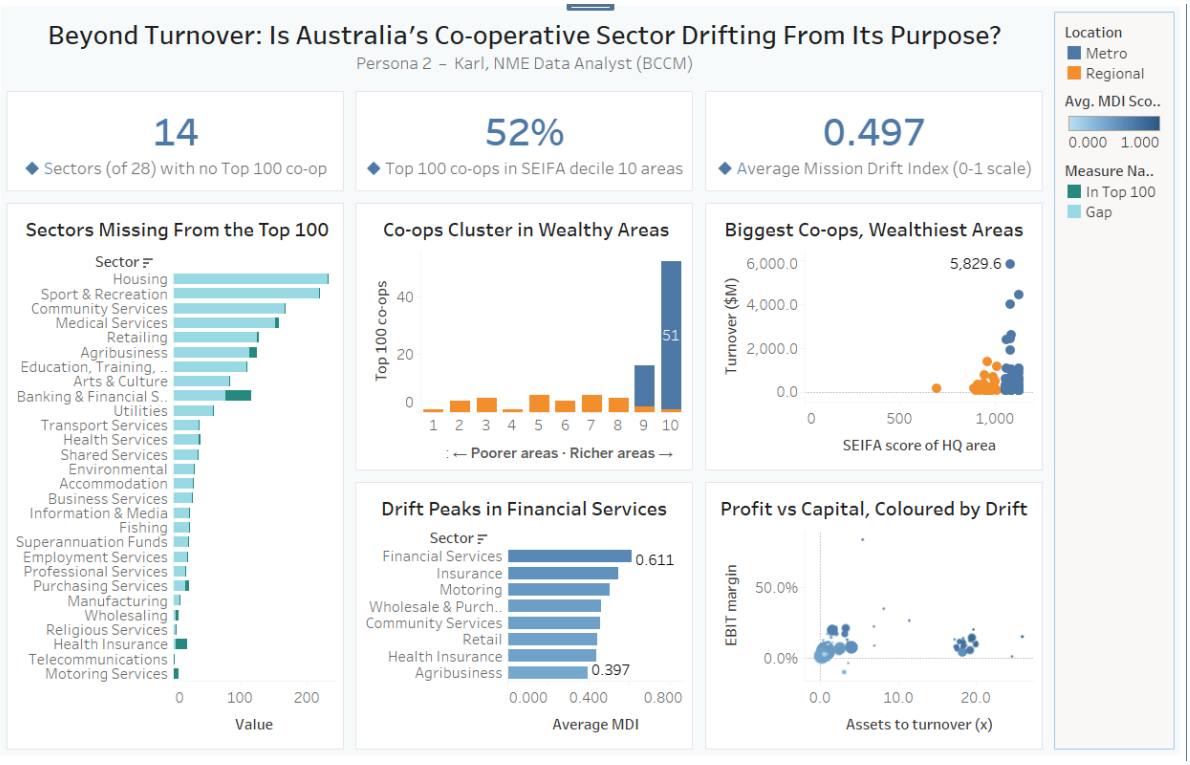


Figure 2: The Persona 2 dashboard in Tableau, showing all three questions.

4.2.1 Question 1: Which sectors show the largest gap between active CMEs and Top 100 representation?

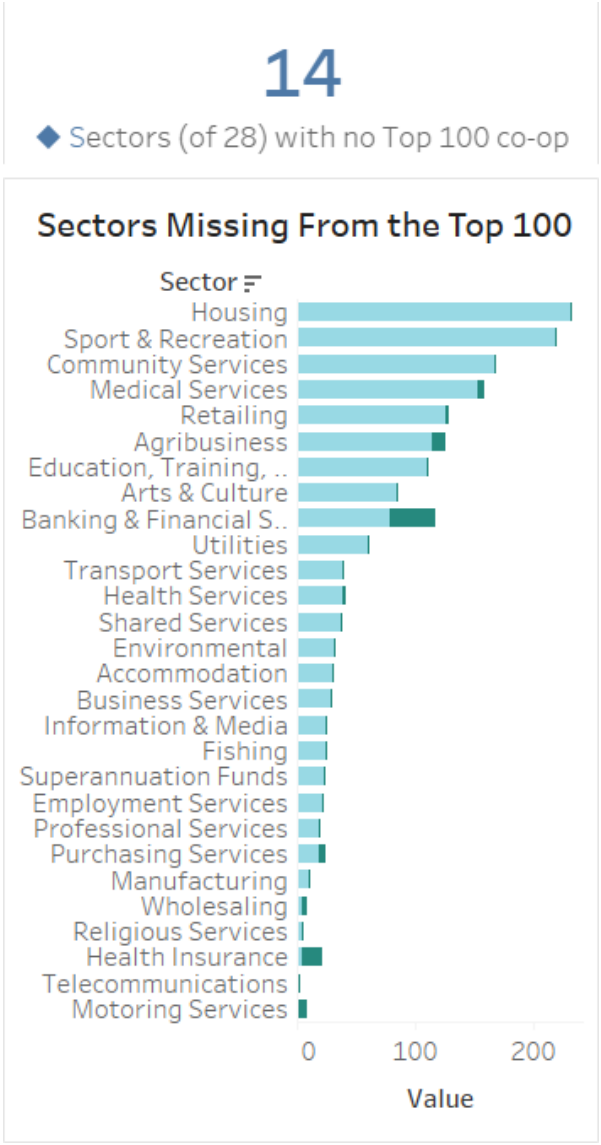


Figure 3: Active CMEs compared with Top 100 seats across the 28 CEMI sectors.

The blind spot. Australia has 1,810 active CMEs across 28 industry sectors, but the NME Report's Top 100 only covers 14 of those sectors. Thirteen sectors are missing, and superannuation is excluded by design. Housing (232 active CMEs) and Arts & Culture (85) have zero representation, and Sport & Recreation (220) has just one seat. These are not small, dying sectors. Housing is the single largest sector by entity count. They are invisible because the Top 100 is ranked by turnover, and community-serving co-ops do not generate the revenue that financial institutions do.

Meanwhile, Health Insurance has 21 active CMEs and holds 17 Top 100 spots, an 81% capture rate. Financial Services has 118 active CMEs and holds 39 spots. The Top 100 is not a picture of the co-operative sector. It is a picture of which co-ops make the most money. For a movement founded on community purpose, that distinction matters.

Decision for Karl. Add a Sector Representation view to the 'Economic impact by industry' section of the NME Report. That section already counts Top 100 co-ops by industry but lists only industries that hold a seat, so extending it to all 28 sectors is a small change that fits the report's existing format. A note should explain that superannuation funds are excluded by design.

4.2.2 Question 2: Are Australia's largest co-ops drifting from the communities that need them most?

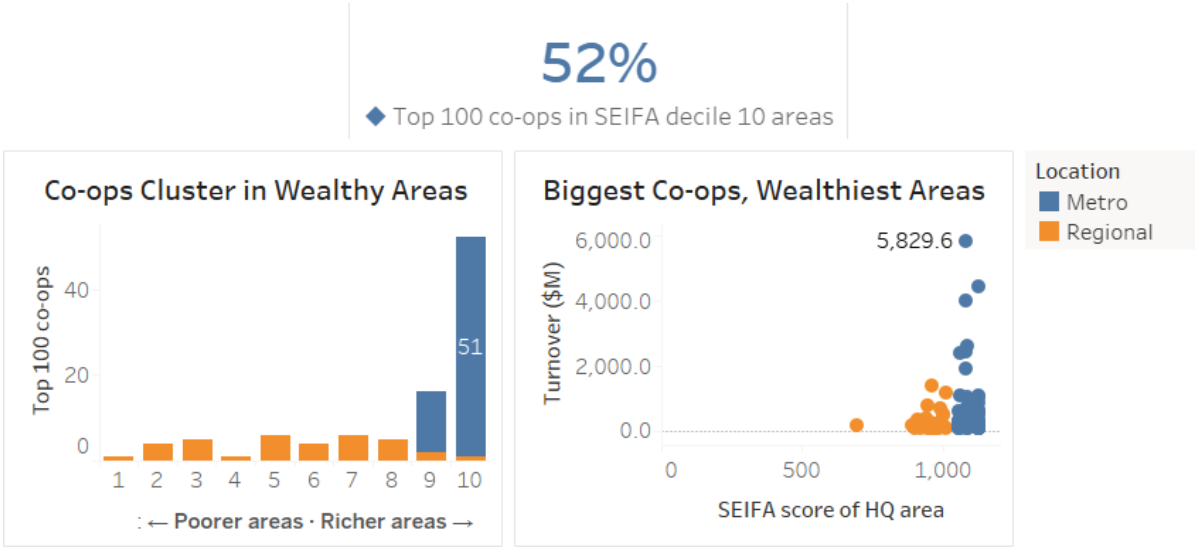


Figure 4: Top 100 head offices by SEIFA decile, and turnover against SEIFA score.

The geographic evidence. Q1 showed which sectors are missing. Q2 asks: where are the co-ops that did make the Top 100 actually located, and who do they serve?

We matched all 100 Top 100 co-op headquarters to ABS SEIFA socio-economic data at the Local Government Area level. The result was stark: 52 of 100 are in SEIFA Decile 10, the most advantaged areas in the country. The average SEIFA score across the Top 100 is 1,043, 43 points above the ABS national average of 1,000. Only 10 co-ops sit in Decile 1 to 3 areas.

The exceptions tell the real story. ALPA (East Arnhem, Decile 1), Kimberley Aboriginal Medical Services (Broome, Decile 8), and Central Australian Aboriginal Congress (Alice Springs, Decile 8) are among the few Top 100 co-ops genuinely serving disadvantaged communities. The Broome and Alice Springs area averages hide the disadvantage these two services exist to address. All three sit in the lower half of the Top 100 by revenue. The co-ops that serve the people who need co-operation most are in the lower half of the leaderboard.

Decision for Karl. Add a Community Impact Overlay showing totals by SEIFA decile, not individual co-ops, so the report shows where co-operative capital concentrates without singling out any member or area. Because this data shows head offices and not where members live, Karl should also start collecting member postcodes to test the result.

4.2.3 Question 3: How far is success pulling Australia's biggest co-ops away from the people who need them most?

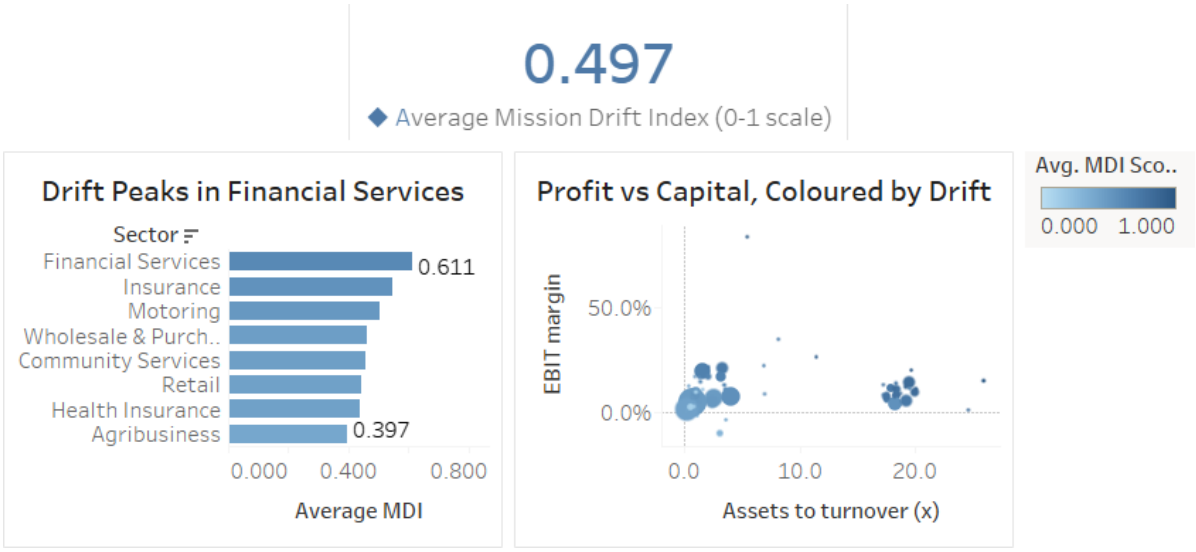


Figure 5: Average MDI by sector, and EBIT margin against assets-to-turnover by co-op.

The measurement. Q1 found the blind spot. Q2 mapped the drift. Q3 asks the hardest question: can we put a number on it? If BCCM wants to act on this problem, Karl needs a metric, not an anecdote.

We constructed a Mission Drift Index (MDI), a composite score from 0 (fully mission-aligned) to 1 (fully drifted), built from three z-score normalised components:

- **Profit Extraction Rate:** EBIT / Turnover. Are members' fees being extracted as profit or returned as value?
- **Capital Hoarding Ratio:** Total Assets / Turnover. Is the co-op circulating wealth or accumulating it?
- **Community Distance:** SEIFA Decile of headquarters. How far is the co-op from the communities that need it?

The MDI draws on established composite index methodology. Mersland and Strøm (2010) measured mission drift in microfinance institutions using average loan size as a proxy, finding that average loan size had not increased across the industry as a whole. Staessens et al. (2019) advanced this further by constructing a composite performance score for social enterprises using Data Envelopment Analysis, enabling simultaneous measurement of social and economic output. Our approach adapts the logic of the Altman Z-score (1968), the foundational composite financial index that combines five accounting ratios into a single bankruptcy risk score. Where Altman's Z-score diagnoses financial distress, our MDI diagnoses mission distress: the same z-score normalisation technique applied to a fundamentally different question.

The results confirmed what Mondragon and REI warned us about. Financial Services co-ops have the highest average MDI (0.611), followed by Insurance (0.545) and Motoring (0.504). Individual scores across co-ops with complete data range from 0.266 to 0.715.

The MDI is not a judgement. A high MDI does not mean a co-op is failing. It means the co-op's financial profile has diverged from the community-serving model the sector was built on. That divergence may be necessary, even healthy. But it should be visible. Right now, it is not.

Decision for Karl. Pilot the MDI internally for two NME reporting cycles before deciding whether to publish it. Two cycles give the first year-on-year comparison, so the index can begin to show change over time, and they give members time to review what their scores mean.

5. Data Quality, Ethics and Do No Harm: Persona 2

5.1 Data quality

We reviewed the data set at the syntactic, semantic and pragmatic levels of data quality (Price & Shanks, 2005).

Syntactic. Fields follow one format per column, and a Metadata sheet defines each one. Every record was fact-checked against its source document. Two records have values missing from the source, one with no EBIT and one with neither EBIT nor assets. We left these blank instead of estimating them, so those two MDI scores use fewer components and should be read with care.

Semantic. The word 'sector' has two meanings in our data: the 28 CEMI sectors in Q1 and 8 project groupings in Q3. This is why Health Insurance has 17 Top 100 seats in Q1 but 19 records in Q3, so the two charts should not be compared directly. The NME and CEMI tables also disagree on which of two organisations ranks 64 and which ranks 65, although their financial values agree. One insurer reports EBIT equal to 84% of turnover. This matches the source but is an extreme outlier, and the index caps it.

Pragmatic. The data must fit the decision. It covers one financial year, so it shows where co-ops are now and cannot on its own show change over time. It also records head offices, not where members live. BCCM should treat the SEIFA and MDI views as prompts for further inquiry, and should check any finding against member-level data and earlier NME years before relying on it.

5.2 Data ethics: PAPA framework

We used the PAPA framework (Mason, 1986) to review the project, and the main issues are summarised below.

Dimension	Issues in this project and how we handled them
Privacy	We collected organisation-level data only, with nothing about individual members or employees. Harm can still occur at group level, because labelling a named co-op or area can stigmatise it. The MDI is therefore reported by sector, with no ranking of individual co-ops as 'most drifted'.
Accuracy	An error in a composite score could damage a member's reputation. We fact-checked every record, documented the formula and disclosed the missing values. BCCM should name one person accountable for the index and let co-ops correct their data before any release.
Property	All data comes from public reports and is used for non-commercial purposes with full acknowledgement. Fields we created, including the MDI, are identified as derived so they are not mistaken for figures published by BCCM, CEMI or the ABS.

Accessibility	Most of the 1,810 active CMEs have no published financial data, so they cannot be scored and have no voice in this view. Terms like z-score need plain explanation for non-technical readers. Sharing the workbook publicly would expose co-op-level scores, so access should start with BCCM staff.

5.3 Visual ethics

What we did. The bar charts start at zero, so bar lengths can be compared fairly. Metro and Regional use contrasting blue and orange, and the drift scale uses one colour running from light to dark on a fixed 0 to 1 scale, which avoids implying a good or bad judgement through red and green.

What to watch for. The SEIFA axis on the turnover chart starts at zero, which squeezes the points to the right, and one very large co-op stretches the vertical axis. Karl should use tooltips to read individual points. Titles such as 'Co-ops Cluster in Wealthy Areas' and the 'Poorer areas' and 'Richer areas' labels are shorthand: SEIFA measures relative socio-economic advantage and disadvantage in an area, not the wealth of a co-op or its members. Wholesale & Purchasing has only one record, so its bar is not a true sector average. Scores shown to three decimals can also look more exact than they are, so small differences should not be treated as meaningful.

Advice on time filters. The data covers one year today. When earlier NME editions are added, a time filter could be used to choose a start year that flatters a trend, or to include or hide unusual years. We advise showing the full available period by default and stating the period in every chart title.

5.4 'Do No Harm' principles

We applied the Do No Harm Guide (Schwabish & Feng, 2021) as follows.

Acknowledge who is missing. Q1 exists to make the sectors outside the Top 100 visible, and the report states that superannuation funds are excluded by design so that this gap is not misread.

Think about ordering. The Q1 chart is sorted by the size of the gap, not by turnover, so the eye goes first to the least represented sectors and not the wealthiest.

Choose language carefully. Terms such as 'Capital Hoarding' describe a financial ratio, not a co-op's motives, and a high MDI is not a judgement. BCCM should review this wording with members before any wider use.

Give context. The ABS warns that area indexes misclassify many people when used as proxies for individuals (ABS, 2023b). Kimberley Aboriginal Medical Services and Central Australian Aboriginal Congress both sit in Decile 8 because of their area averages, although they serve communities facing disadvantage, and the Alice Springs service's MDI of 0.504 is close to the average. These scores should not be read as drift. Any use of data about Aboriginal community-controlled organisations should involve them and follow Indigenous data governance principles (Maiam nayri Wingara Collective, 2018).

6. Recommendations and Future Work: Persona 2

The NME Report is the most widely cited document on Australia's co-operative sector. It is how policymakers, regulators, and the public understand what co-ops contribute. But by measuring the sector exclusively through revenue, it inadvertently tells a story where the biggest co-ops are the best co-ops, and the community-serving ones are invisible.

We recommend three additions to the NME Report framework:

1. **Sector Representation Dashboard:** Show all 28 sectors and their Top 100 visibility, so the report acknowledges what the revenue lens hides.
2. **Community Impact Overlay:** Map Top 100 co-ops against SEIFA data to show where co-operative capital concentrates and where it is absent.
3. **Mission Drift Index (pilot):** Introduce MDI as a supplementary measure alongside turnover, giving BCCM an early warning system for purpose erosion across the sector.

These additions do not replace what the NME Report already does well. They complement it, so the report can tell the full story of Australia's co-operative sector, not just the story of its wealthiest members.

Future use and management. BCCM should nominate one staff member to own the dashboard and refresh it once a year when NME data is finalised, repeating the same fact check against the source documents. Access should start with BCCM staff. Sector-level views can be shared with members and the public after the pilot, while co-op-level MDI scores stay internal.

Further development. The first priority is adding earlier NME editions, so Q3 can measure change over time. Next is member postcodes, so Q2 can show who co-ops serve as well as where they are based. BCCM could then consider ORIC data on Aboriginal and Torres Strait Islander corporations, the ACNC charity register for social outcomes, and APRA statistics for mutual banks and insurers.

7. References: Persona 2

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